Workout 1B Homework 2 Challenge

Below is a scored review of your assessment. All questions are shown.

Earned 79,992 of 99,990 points. **80%**

Main Question Set

Solve six of the first ten questions to achieve a passing score of 60%. Don't miss the learning opportunity in the rest of the problems, though -- spend the full time and attempt every one of them!

1 In the sequence which starts with the terms $1, 3, 6, 10, 15, 21, \ldots,$ the nth term is equal to what?

Explanation

It's easy to see that this sequence grows too slowly for each term to be equal to the sum of the first n squares or cubes. By comparison, those two sequences are $1, 5, 14, 30, 55, \ldots$ and $1, 9, 36, 100, 225, \ldots$. The sum of the first n odd numbers is equal to n^2, but the terms in this sequence aren't squares, so that disqualifies that choice. If the terms were the sum of the first n even numbers, then every term should be even, but we have odd terms in our sequence $(1, 3, 15, 21\ldots)$. The remaining choice is that the sequence represents partial sums of the numbers from 1 to n, and this works if you recall that $1 + 2 + 3 + \ldots + n = \frac{n(n+1)}{2}$, which you can derive by simply adding $1 + 2 + 3 + \ldots + n$ to itself, forming n groups of $(n + 1)$, and then dividing by 2 to get the original sequence again. The numbers in this sequence, $1, 3, 6, 10, \ldots,$ are called **triangular numbers**, since they can be drawn as dots arranged in a triangle formation, with each row having one more dot than the previous row. For example, the 6th triangular number, 21, is depicted below. The answer is $\boxed{\text{the sum of the numbers from 1 to } n}$.

$$\bullet = 1$$
$$\bullet\ \bullet = 2$$
$$\bullet\ \bullet\ \bullet = 3$$
$$\bullet\ \bullet\ \bullet\ \bullet = 4$$
$$\bullet\ \bullet\ \bullet\ \bullet\ \bullet = 5$$
$$\bullet\ \bullet\ \bullet\ \bullet\ \bullet\ \bullet = 6$$

$$\text{total } = 1 + 2 + 3 + 4 + 5 + 6 = 21$$

2

Suppose we have a set with elements $\{0, -1, -2, -2, -3, -3, -3, 4, 4, 4, 4, 5, 5, 5, 6, 6, 7\}$. Which of the following will change if we replace each element x in the set with $|x|$? Choose all that apply.

Explanation

The elements in this set happen to conveniently be already ordered by increasing absolute value. In order to determine the original median, the original sequence should start with $-3, -2, -2, -1, \ldots$. Then, after replacing each value with its absolute value, the negative numbers move to the right of the 0 and take on the order shown in the problem. The median, the middle number, 4, will stay the median. Also, the most frequent number (mode), 4, will still be the most frequent number. The sum of the numbers is bigger than before, since the negatives became positive, so the mean changes. The range, which used to be $7 - (-3) = 10$, is now only $7 - 0 = 7$, so it changes as well. (It's easy to be fooled into thinking the smallest number before was 0, so the range used to be $7 - 0$ as well.) The answer is $\boxed{\text{mean and range}}$.

3 Which of the following is closest to the value of $\log_{10} \pi$?

Explanation

We saw in the Day 7 Growing by Pi Part 1 lesson that π is approximately equal to $\sqrt{10}$, so $\log \pi \approx \log 10^{\frac{1}{2}} = \boxed{\frac{1}{2}.}$

4 What is the biggest prime factor of $(22,206)^2-(5,822)^2$?

Explanation

This is a difference of squares, and we can write this as $(22206+5822)\,(22206-5822)$, which turns into $(28028)(16384)$. Which one has the larger prime factor, 28028 or 16384? The pattern of digits in 28028, which has 28 shifted over by three digits and added to itself, is a big sign that it has 1001 as a factor. Recall from the Module 1 Day 7 or Module 1 Day 13 lessons that 1001 can be prime factorized as $1001=7\times 11\times 13$. So $28028=28(1001)=(28)(7)(11)(13)$, and the largest prime factor of 28028 is 13. Now to compare with 16384. We saw this in the Day 4 (Double Factorial Part 3) lesson! This is 2^{14}, so its biggest prime factor is just 2. Thus we know that the largest prime factor of $22,206^2-5,822^2$ is $\boxed{13}$.

5 What is the 1.2^{th} root of 64?

Explanation

Let x equal the 1.2^{th} root of 64. Then:

$$x^{1.2} = 64$$
$$x^{\frac{12}{10}} = 64$$

$$x^{\frac{6}{5}} = 64$$

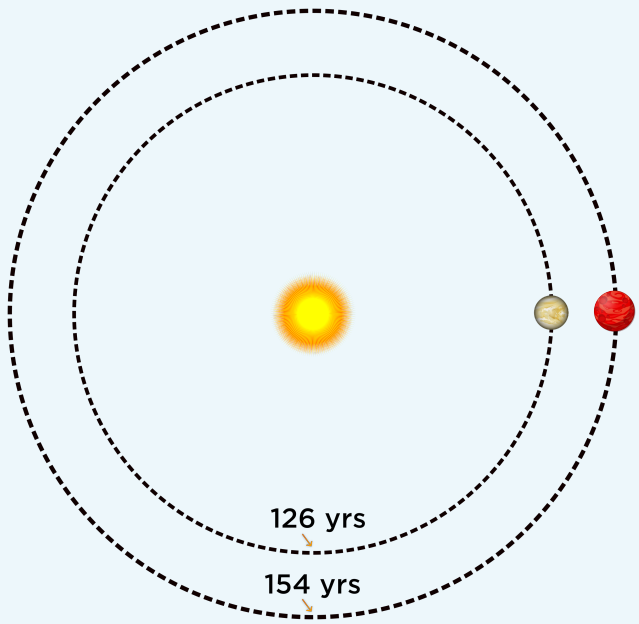
$$\sqrt[6]{x^{\frac{6}{5}}} = \sqrt[6]{64}$$

$$x^{\frac{1}{5}} = 2 \qquad (\text{Since } 64 = 2^6)$$

$$\left(x^{\frac{1}{5}}\right)^5 = 2^5$$

$$x = \boxed{32}.$$

6 Two planets are orbiting a star, with orbit lengths of 154 years and 126 years. Right now, the planets are lined up to the right of the star. After how many years will the two planets be lined up at this position again?



Explanation

The planets start together, so by the time they are side-by-side again, they must have each completed an integer number of orbits. (Since the question indicates they should return to the same position, this question is less tricky than the one from the Day 8 (Orbits Matching Up) lesson, since we don't have to worry about them lining up at partial orbit lengths.)

We can express each planet's speed in terms of $\frac{\text{cycles}}{\text{year}}$, so the first planet goes at $\frac{1}{154}\frac{\text{cycles}}{\text{year}} = \frac{1}{7\times2\times11}\frac{\text{cycles}}{\text{year}}$. The second planet goes at $\frac{1}{126}\frac{\text{cycles}}{\text{year}} = \frac{1}{7\times2\times3^2}\frac{\text{cycles}}{\text{year}}$. To find distance in cycles, multiply speed with time.

$$(\text{distance}) = (\text{speed}) \times (\text{time}).$$

In time t, the first planet will have traveled $\left(\frac{1}{7\times2\times11}\frac{\text{cycles}}{\text{year}}\right)(t \text{ years}) = \frac{t}{2\times7\times11}$ cycles. The second planet will have traveled $\left(\frac{1}{2\times3^2\times7}\frac{\text{cycles}}{\text{year}}\right)(t \text{ years}) = \frac{t}{2\times3^2\times7}$ cycles. In order for both distances to be a whole number of orbits, t should equal the LCM of $2\times7\times11$ and $2\times3^2\times7$, which is $2\times3^2\times7\times11 = \boxed{1386}$ years.

7 Alan lives on the odd-numbered side of his street. Let a equal the number of houses to the left, in the increasing direction, next to Alan's house, and let x equal the number of Alan's house. In terms of a and x, what is the sum of the house numbers to the left of Alan's house, not including Alan's house?

Explanation

The houses to Alan's left are $x+2, x+4, x+6, \ldots$, with a houses in total. The 1st house is $x+2$, the 2nd house is $x+4$, the 3rd house is $x+6$, and so on, so the nth house is $x+2n$. That means the last house is $x+2a$. We want the sum $x+2, x+4, x+6, \ldots + x+2a$. If we add this to itself, we can pair $x+2$ with $x+2a$, and pair $x+4$ with $x+2a-2$, and so on, so that we get a pairs, each summing to $2x+2a+2$. Multiplying this out, we get $a(2)(x+a+1)$, but we need to also divide by 2 since we added the series to itself. The final sum is $\boxed{a(x+a+1)}$.

8 Elena is standing in the middle between two houses, on one side of a street full of houses. The house numbers on any side are either all even or all odd. If she totals the numbers of the houses on either side of her, the total is the same. What is the smallest possible average of the two house numbers on either side of Elena?

Explanation

Let the average of the two house numbers be x. Suppose there are a houses to the left of Elena, numbered $x-1, x-3, x-5, \ldots$. To the right of Elena, there are b houses, numbered $x+1, x+3, x+5, \ldots$. The sum of the house numbers on the left is $x-1+x-3+x-5+\ldots+x-(a^{\text{th}} \text{ odd })$, and the sum of the house numbers on the right is $x+1+x+3+x+5+\ldots x+(b^{\text{th}} \text{ odd })$. Let's use the fact that the sum of the first n odds equals n^2. Then we have:

$$\begin{aligned} \text{sum of numbers on left} &= \text{sum of numbers on right} \\ ax-a^2 &= bx+b^2 \\ (a-b)x &= a^2+b^2 \end{aligned}$$

$$x=\frac{a^2+b^2}{a-b}$$

Analyzing this shows that as a and b grow larger, the value of x also grows larger, since the square of a number grows faster than the number. The minimum value occurs when $a=2$, $b=1$, which gives $x=\frac{2^2+1^2}{2-1}=\frac{5}{1}=\boxed{5}$.

9 What number comes next in this sequence: $3, 8, 24, 48, 120, 168, ?$

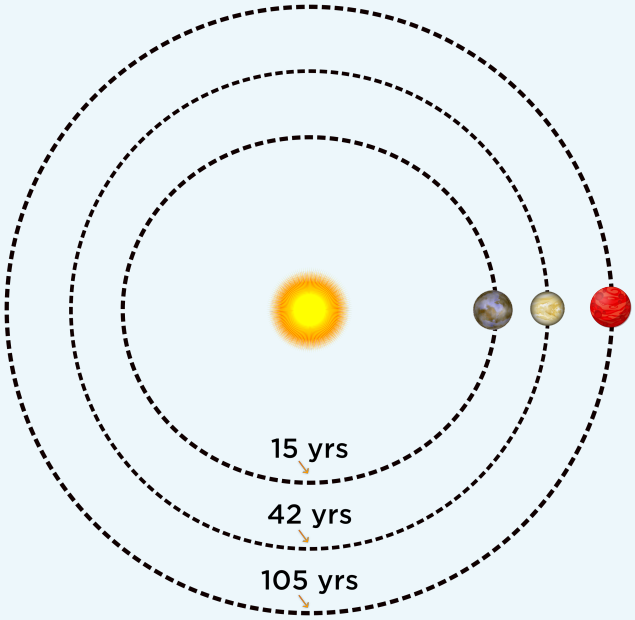
Explanation

Explanation: This sequence will not be described well using a common difference or common ratio (especially since the jump from 48 to 120 is larger than from 120 to 168 using either metric). The important realization is that all of the listed numbers are one less than a perfect square. The terms can be rewritten as

$$2^2 - 1,\ 3^2 - 1,\ 5^2 - 1,\ 7^2 - 1,\ 11^2 - 1,\ 13^2 - 1$$

This expression makes the pattern apparent; the next term of the sequence should be one less than the square of the next largest prime number. The first prime after 13 is 17, so the next number should be $17^2 - 1 = 289 - 1 = \boxed{288}$.

10 Three planets are orbiting a star, with orbit lengths of 105 years, 42 years, and 15 years. Right now, the planets are all lined up in a perfect eclipse. After how many years will the planets and star be collinear again, in the sense that there is a straight line that goes through all of them?



Explanation

As Prof. Loh showed us in the Day 8 Orbits Matching Up Part 2 lesson, a good idea is to pretend that the slowest planet is staying in place, while the other planets are moving, and adjust their speeds accordingly. Then we only have to worry about matching up two of the planets' positions. The planets, from slowest to fastest, go at speeds $\frac{1}{105}\frac{\text{cycles}}{\text{year}}$, $\frac{1}{42}\frac{\text{cycles}}{\text{year}}$, and $\frac{1}{15}\frac{\text{cycles}}{\text{year}}$. The speeds of the planets relative to the slowest planet are as follows:

$$\text{Speed}_{\text{slowest}} = 0 \qquad \frac{\text{cycles}}{\text{year}}$$

$$\begin{aligned}
\text{Speed}_{\text{middle}} &= \frac{1}{42} - \frac{1}{105} && \frac{\text{cycles}}{\text{year}} \\
&= \frac{1}{3 \times 2 \times 7} - \frac{1}{3 \times 5 \times 7} \\
&= \frac{5}{3 \times 2 \times 5 \times 7} - \frac{2}{3 \times 2 \times 5 \times 7} \\
&= \frac{3}{3 \times 2 \times 5 \times 7} \\
&= \frac{1}{2 \times 5 \times 7} && \frac{\text{cycles}}{\text{year}}
\end{aligned}$$

$$\begin{aligned}
\text{Speed}_{\text{fastest}} &= \frac{1}{15} - \frac{1}{105} && \frac{\text{cycles}}{\text{year}} \\
&= \frac{1}{3 \times 5} - \frac{1}{3 \times 5 \times 7} \\
&= \frac{7}{3 \times 5 \times 7} - \frac{1}{3 \times 5 \times 7} \\
&= \frac{6}{3 \times 5 \times 7} \\
&= \frac{2}{5 \times 7} && \frac{\text{cycles}}{\text{year}}
\end{aligned}$$

With the slowest planet as the anchor point, the planets will line up when the faster two planets have traveled integer or integer-and-a-half orbit cycles. Looking at the denominators of the speeds of the faster two planets, and using the fact that $(\text{distance}) = (\text{speed}) \times (\text{time})$, we want to multiply a time t to the speeds $\frac{1}{5\times7\times2}$ and $\frac{2}{5\times7}$ so that the denominators are either 2 or 1. We see that the smallest t that works is 35 years, since then the faster two planets will have gone $\frac{1\times5\times7}{5\times7\times2} = \frac{1}{2}$ and $\frac{2\times5\times7}{5\times7} = 2$ orbit cycles, respectively. (Keep in mind these aren't the actual orbit distances, but the adjusted ones in this other frame of reference where the slowest planet is still.) Thus the answer is $\boxed{35}$ years.

Earned 0 of 7 points. 0%

Bonus Challenge Questions

These optional questions will not count towards your final score. However, we strongly suggest trying them for the benefit of the challenge!

11 Suppose there is a three-digit number abc such that the number equals $a + b^2 + c^9 - c^2 - c$. Which of the following is a possible value for c?

Explanation

We know that a, b, and c are single-digit integers. We also have a ninth power (c^9) which can't get too big, since the expression is a three-digit number. Remembering some numbers off the top of our head, we might know that $9^3 = 729 = (3^2)^3 = 3^6$, so 3^7 will exceed 1000 and then 3^9 will definitely be bigger than a 3-digit number. Thus c is less than 3, so it can only be 1 or 2. If $c = 1$, then the biggest possible value is when $a = b = 9$, giving $a + b^2 + c^9 - c^2 - c = 9 + 9^2 - 1 = 89$, which isn't a three-digit number. Thus the only possibility is $c = \boxed{2}$.

12 Which of the following is equal to $\frac{1000}{996}$?

Explanation

As we saw in the Day 5 (Fraction to Decimal) video lesson, the trick is to split off 1, leaving $\frac{4}{996}$ remaining, which we express as $\frac{4}{1000}$ with some correction factor. We do this successively, which allows us to determine the digits of the number. The first step is to write the following:

$$\frac{1000}{996} = \left(1 + \frac{4}{996} + \dots\right)$$

Since we like fractions with denominators like 10^n, it makes sense to replace $\frac{4}{996}$ with $\left(\frac{4}{1000}\right)\left(\frac{1000}{996}\right)$, the latter being a correction factor.

$$\frac{1000}{996} = \left[1 + \frac{4}{1000}\left(\frac{1000}{996}\right)\right]$$

Continuing this pattern, we peel off another 1 to get the next term, like so:

$$\frac{1000}{996} = \boxed{1 + \frac{4}{1000}\left(1 + \frac{4}{996}\right)}.$$

13 Using only a single 1 and a single 2, and any operation including $+, -, \times, \div$, along with using powers, factorials or decimals, what is the maximum value you can get? (Note: adding a 0 isn't allowed.)

Explanation

Let's see, it's possible to get 2 by doing 1×2, but $1 + 2 = 3$ is bigger. We can get even bigger, though, by dividing by a decimal. Between $\frac{2}{.1}$ and $\frac{1}{.2}$, the first is the bigger one, equal to $\frac{2}{.1} = \frac{20}{1} = 20$. But can we do better? We can! If we use the 2 as a power instead of the numerator, it will have a bigger effect. Oh, but $\left(\frac{1}{.1}\right)^2$ uses an extra 1 for the numerator. No problem! The solution is to use a negative power. Then we have $.1^{-2}$, which is the same as $\left(\frac{1}{.1}\right)^2 = \boxed{100}$.

14 What is the value of

$$\log_{10}\left[3(1^2)+3(2^2)+3(3^2)+3(4^2)+3(5^2)+3(6^2)+3(7^2)+3(8^2)+3(9^2)+3(10^2)\right.$$
$$\left.-1-3-5-7-9-11-13-15-17-19-55\right]?$$

Explanation

You don't have to add up the terms by hand here -- just spot a few patterns and employ some formulas that we saw previously. In particular, $3(1^2)+3(2^2)+3(3^2)+3(4^2)+3(5^2)+3(6^2)+3(7^2)+3(8^2)+3(9^2)+3(10^2)$ equals 3 times the sum of the first 10 squares. Recall the formula which we saw in the Week 1 Challenge, which says:

$$1^2+2^2+3^2+4^2+5^2+\ldots n^2=\frac{n(n+1)(2n+1)}{6}$$

$$3\left(1^2+2^2+3^2+4^2+5^2+\ldots n^2\right)=\frac{n(n+1)(2n+1)}{2} \qquad \text{(multiplying both sides by 3)}$$

$$=\frac{2n^3+3n^2+n}{2}$$

$$=n^3+\frac{3n^2}{2}+\frac{n}{2}.$$

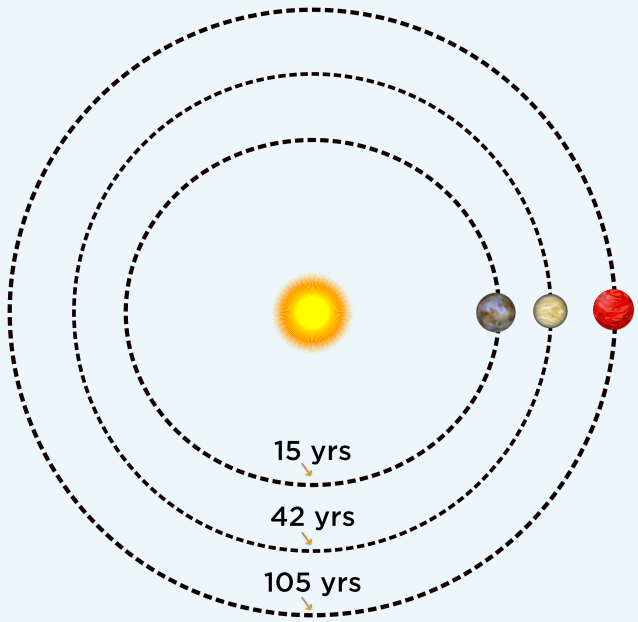
Putting this back into the original expression, we want to find:

$$\log 10\left[10^3+\frac{3}{2}\cdot 10^2+\frac{10}{2}-1-3-5-7-9-11-13-15-17-19-55\right]$$

The expression $1+3+5+7+9+11+13+15+17+19$ is the sum of the first 10 odd numbers, which we saw in the Day 7 Growing by Pi Part 4 lesson, equals 10^2.

$$\log 10\left[10^3+150+\frac{10}{2}-10^2-55\right]$$
$$=\log_{10}\left[10^3\right]$$
$$=\boxed{3}.$$

15 Three planets are orbiting a star, with orbit lengths of 105 years, 42 years, and 15 years. Right now, the planets are all lined up in a perfect eclipse. After t years, the planets and star will be collinear again, in the sense that there is a straight line that goes through all of them. How many orbits will the slowest planet and fastest planet have gone after time t? Your answers should be fractions in the form $\frac{a}{b}$, where a and b are integers with no common factors.



Slowest planet: _1._ / _2._ orbits

Fastest planet: _3._ / _4._ orbits

Explanation

Let's not make the mistake of using the lengths of $\frac{1}{2}$ orbit and 2 orbits from the previous question, because those are based on the frame of reference of the slowest planet not moving. Instead, we know that the time it takes for the planets to line up again is 35 years. The speeds of the planets, from slowest to fastest, were $\frac{1}{105} = \frac{1}{3\times5\times7}\ \frac{\text{cycles}}{\text{year}}$, $\frac{1}{42} = \frac{1}{3\times2\times7}\ \frac{\text{cycles}}{\text{year}}$, $\frac{1}{15} = \frac{1}{3\times5}\ \frac{\text{cycles}}{\text{year}}$, we can solve for the number of orbits using the distance formula, $(\text{distance}) = (\text{speed}) \times (\text{time})$.

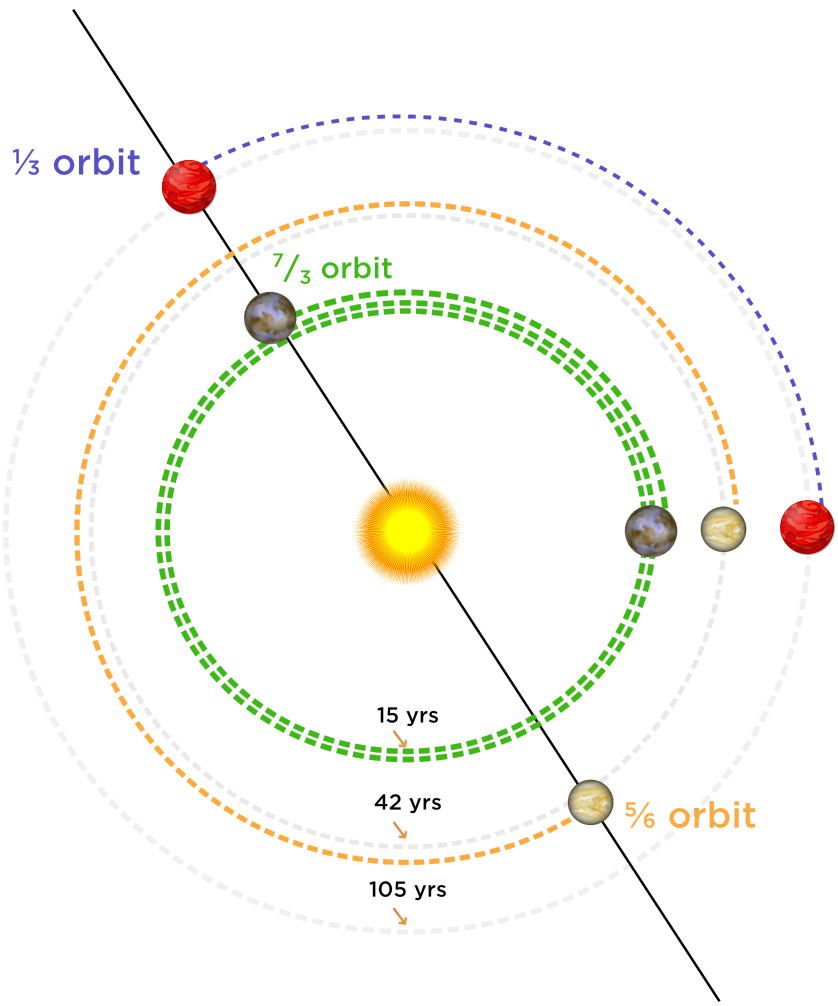
$$\text{Number of orbits}_{\text{slowest}} = \left(\frac{1}{105}\,\frac{\text{orbits}}{\text{year}}\right)(35\text{ yrs})$$

$$= \boxed{\frac{1}{3}}\text{ orbits}$$

$$\text{Number of orbits}_{\text{fastest}} = \left(\frac{1}{15}\,\frac{\text{orbits}}{\text{year}}\right)(35\text{ yrs})$$

$$= \boxed{\frac{7}{3}}\text{ orbits.}$$

The planets would be lined up as in the diagram below.



16 What number comes next in the sequence $1, 4, 3, 11, 15, 13$?

Explanation

No common sequence analysis tactic works with this sequence. However, one can note the increasing length of the written form of each number:

1	4	3	11	15	33
One	Four	Three	Eleven	Fifteen	Thirteen
3	4	5	6	7	8

In fact, each number is the smallest positive integer with a certain number of letters. The next number should be the smallest positive integer with nine letters in its written expression, and that number is $\boxed{17}$ (seventeen).

17 Convert the fraction $\frac{10}{9996}$ into a decimal. Find the sum of the first five *non-zero* digits of this decimal. (It's not zero!)

Explanation

As we saw in a previous question, the trick here is to approximate this based on a close fraction, $\frac{10}{10000}$, and split off more fractions like $\frac{10}{9996}$ to create an infinite geometric series. The idea is to create a lot of fractions with denominators in the form 10^n, so that it's easy to see the digits of the decimal.

We started with $\frac{1000}{9996} = \left[1 + \frac{4}{10000}\left(\frac{10000}{9996}\right)\right]$ and continued this pattern like so:

$$\begin{aligned}\frac{10000}{9996} &= 1 + \frac{4}{10000}\left(1 + \frac{4}{9996} +\right) \\ &= 1 + \frac{4}{10000}\left(1 + \frac{4}{10000}\left(1 + \frac{4}{10000}\left(1 + \frac{4}{9996}\right)\right)\right)\end{aligned}$$

The nonzero terms will come from the powers of $\frac{4}{10000}$ that come from the expansion, like $\left(\frac{4}{10000}\right)^1, \left(\frac{4}{10000}\right)^2, \left(\frac{4}{10000}\right)^3, \left(\frac{4}{10000}\right)^4$, etc. The digits from these separate powers are offset by at least 4 decimal places, so they won't overlap with each other. In the very front, we still have a 1. Thus the first five non-zero digits come from $1, 4, 16, 64$, and the sum of these digits is $1 + 4 + 1 + 6 + 6 + = \boxed{18}$.

Earned 0 of 3 points. 0%

Super Bonus Questions

These questions are an extreme challenge! Some of the most interesting, creative and clever questions are the hardest ones. So try every single one. Even if you can't figure it out, you'll still be deepening your understanding of the important math concepts that underlie this course.

18 Alan lives on the even-numbered side of his street. If he totals the numbers of the houses on either side of him, the total is the same. If Alan's house number is a two-digit multiple of 10, what is the smallest possible value of Alan's house?

Explanation

Recall from a previous question that the formula for Alan's house number is:

$$x = \frac{b(b+1) + a(a+1)}{a-b},$$

where a equals the number of houses on the side of Alan which has more houses, and b equals the number of houses on the other side of Alan's house. We are interested in products of consecutive numbers, like this:

$$1\times2=2$$
$$2\times3=6$$
$$3\times4=12$$
$$4\times5=20$$
$$5\times6=30$$
$$6\times7=42$$
$$7\times8=56$$
$$8\times9=72$$
$$9\times10=90$$
$$10\times11=110$$
$$11\times12=132$$
$$12\times13=156$$
$$13\times14=182$$
$$14\times15=210$$
$$15\times16=240$$
$$16\times17=272$$
$$17\times18=306$$
$$18\times19=342$$
$$19\times20=380$$
$$20\times21=420$$

Since we want Alan's house number to be a multiple of 10, we are interested in numbers that will add up to a multiple of 10, and this is only possible with numbers that end in 0, since the other numbers end in 2 or 6, which can't add with anything else to make a multiple of 10. The products ending in 0 are shown colored in red.

$$4\times5=20$$
$$5\times6=30$$
$$9\times10=90$$
$$10\times11=110$$
$$14\times15=210$$
$$15\times16=240$$
$$19\times20=380$$
$$20\times21=420$$

A choice of $a=20$ and $b=5$ give the smallest value of $x=\frac{20(21)+5(6)}{20-5}=\frac{450}{15}=\boxed{30}$.

19 Suppose there is a three-digit number abc such that the number equals $a + b^2 + c^9 - c^2 - c$. What is the sum of the possible three-digit numbers?

Explanation

From earlier, we know that $c = 2$, since if $c \geq 3$, then c^9 will be larger than a a three-digit number, and if $c = 1$, then $a + b^2 + c^9 - c^2 - c$ will be smaller than a three-digit number. Using $c = 2$, we can express abc as $100a + 10b + c$ and write:

$$\begin{aligned} 100a + 10b + 2 &= a + b^2 + 2^9 - 2^2 - 2 \\ 100a + 10b + 2 &= a + b^2 + 512 - 2^2 - 2 \\ 99a + 10b &= b^2 + 504 \end{aligned}$$

All three a, b, and c are single digits. The number on the right is a 500's number, so we have that $99a + 90$ is a 500's number, so $a \geq 5$. Also, $512 + b^2$ is less than 600, so $a \leq 6$. If $a = 6$, then b must equal 0, so $a \neq 6$. Thus $a = 5$.

$$\begin{aligned} 99(5) + 10b &= b^2 + 504 \\ 495 + 10b &= b^2 + 504 \\ 10b &= b^2 + 9 \end{aligned}$$

This is true if $b = 1$ or $b = 9$. We will get $10 = 1 + 9$ and $90 = 81 + 9$, respectively. Thus we know that possible solutions are $a = 5, c = 2$, and $b = 1$ or $b = 9$. The two three-digit numbers are 512 and 592, and their sum is $\boxed{1104}$.

20 What number comes next in this sequence: $1, 2, 2, 3, 2, 4, 2, 4, 3, 4, 2, 6, 2, 4, 4, ?$

Explanation

The nth number of the sequence gives the number of positive factors of n. For example, let a_n be the nth term of the sequence. Note that $a_1 = 1$ and $a_n = 2$ when n is prime. The next term of the sequence will be the sixteenth. The factors of 16 are $16, 8, 4, 2,$ and 1, so $a_{16} = \boxed{5}$.

Summary

Software by

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Version 11.2

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